4th **EXCELLENCE IN CONSTRUCTION, DESIGN & ENGINEERING OF EV & BATTERY FACTORIES**
FORUM FOR NORTH AMERICA

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AI Era's Agile Lean Manufacturing in Gigafactory: Lessons learnt from the Battery Giga Factories

Daniel Li
Battery Manufacturing Executive

EV & Battery Manufacturing Experiences

2007-2021: 14 years in Wanxiang America Corp to build Wanxiang Group's global clean energy platform

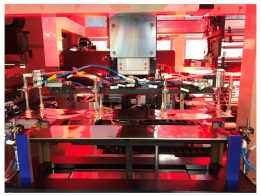
- ✓ Executive VP of Wanxiang Clean Energy USA
- ✓ President of A123 Systems
- ✓ Board of Directors – Karma Automotive

2021-2023: Hyzon Motors

- ✓ Senior VP of Power Solutions
- ✓ President of Hyzon China

2023-2025: SES AI

- ✓ Chief Manufacturing Officer: AI for Manufacturing



2026-: ManufAgent Solutions

- ✓ Managing Director

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Building and Operating a Battery Gigafactory

5M1E Lens

- Man
- Machine
- Material
- Method
- Measurement
- Environment

Success Depends On

- Man – Skilled and safety-compliant workforce
- Machine – Reliable, adaptive equipment
- Material – Stable and traceable supply
- Method – Standardized digital processes
- Measurement – Real-time data visibility
- Environment – Controlled, stable conditions

Gigafactory Challenge

Breakdown in any pillar drives waste, delays, or drift

Man: Skills, Knowledge, Safety

Challenges

- Massive workforce ramp-up amid shortages
- Tacit know-how loss as experts retire
- Safety risks from manual handling and legacy tools
- Bureaucratic inertia in Western operations

Enablers

- Digital training ecosystems: machine teaching
- AI-driven operator guidance: copilot on the floor
- Knowledge capture automation: fast learning
- institutionalized know-how: durable corporate asset

Material: Supply and Quality

Challenges

- Critical mineral volatility (Li, Ni, Co, graphite)
- Supply chain fragility, import reliance
- Traceability gaps across global suppliers

Enablers

- Vertical integration
- Digital material passports
- Predictive procurement & analytics

Machine: Reliability and Flexibility

Challenges

- Immature process lines, reliance on integrators
- Obsolescence amid rapid tech shifts
- Vendor lock-in, limited modularity
- Inconsistent data standards

Enablers

- Commonality, Standardization, Modularization
- Software-defined systems
- Predictive maintenance from unified data

Method: Process and Standardization

Challenges

- Manual or semi-auto bottlenecks
- Complex process
- Non-standardized protocols and formats
- Limited automation ROI in small-scale onshoring

Enablers

- Amplifier: optimization is critical
- Unified digital thread (R&D → Ops)
- AI-assisted scheduling and adaptive SOPs

Measurement: Quality and Data

Challenges

- Fragmented data and disconnected systems
- Slow QA cycles, late defect discovery
- Limited AI readiness, no data historian
- High scrap rates (up to 33%)

Enablers

- Inline quality inspection
- Centralized data lakes: a single source of truth
- Closed-loop feedback systems

Environment: Factory Conditions

Challenges

- Temperature and humidity drift affecting process stability
- Airborne particulate and cleanliness control
- Power or vibration variability degrading yield

Enablers

- Stable HVAC and cleanroom/dryroom control
- Power resilience systems
- Real-time environmental monitoring
- Dynamically adapts manufacturing conditions to save energy.

Towards Gigascale Maturity

Pattern

Misalignment

+

Immaturity

+

Data Fragmentation

=

Scrap & Delay

Success Formula

Align all 5M1E pillars

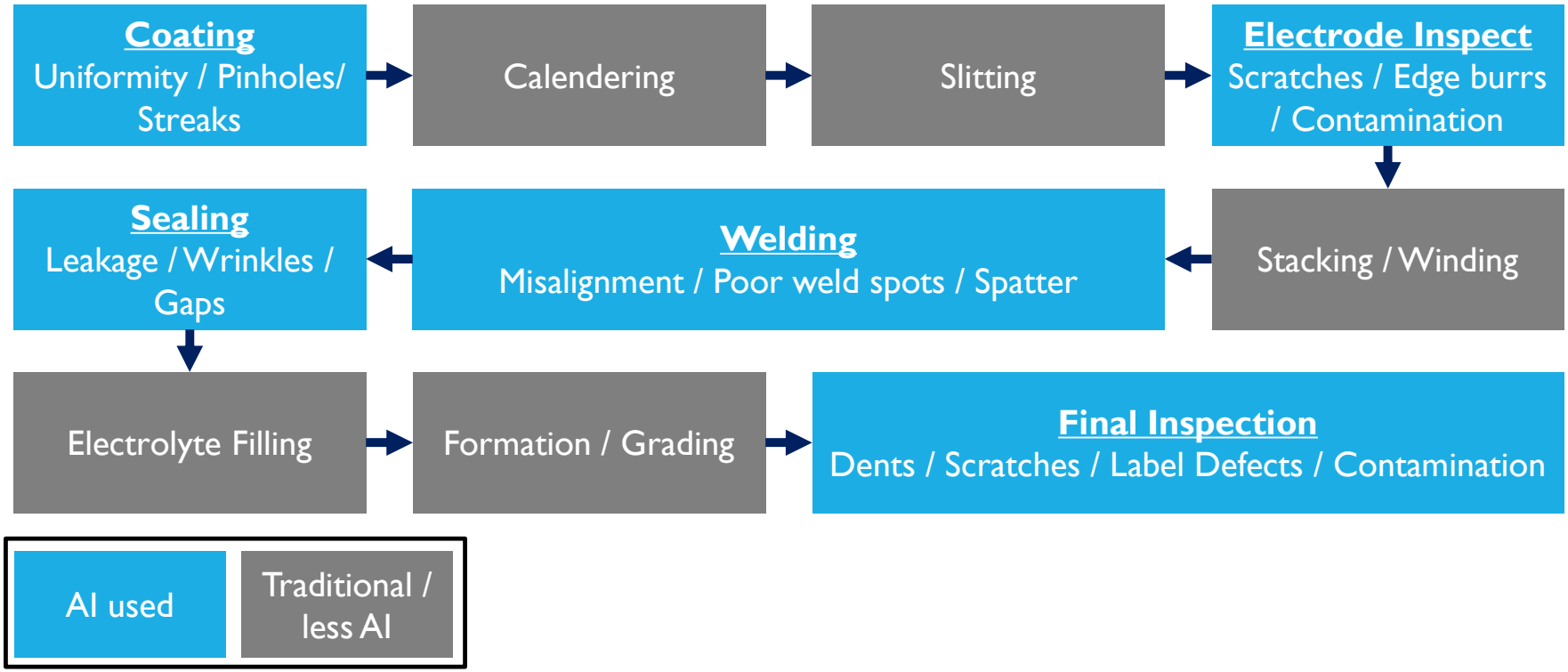
+

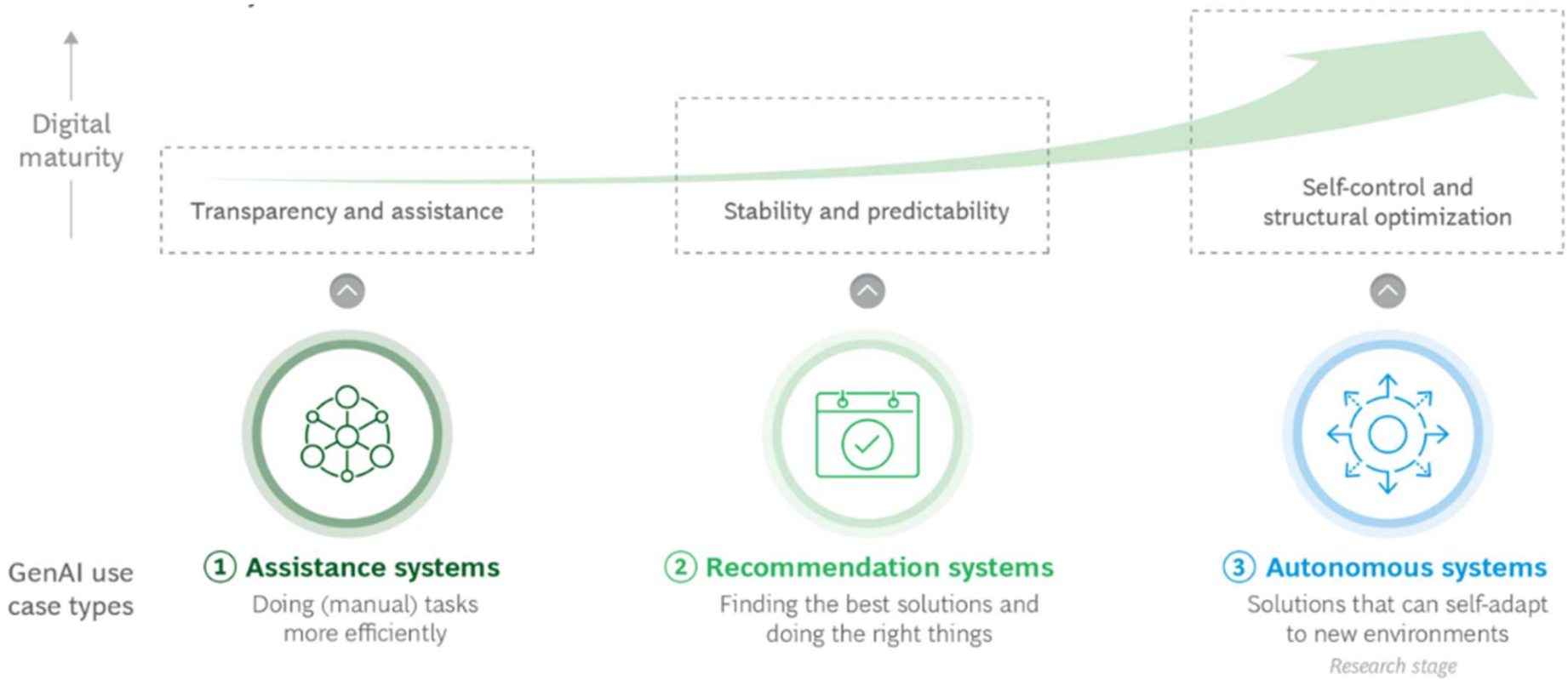
Standardize data and interfaces

+

Evolve into adaptive, digital-first manufacturing systems

AI in Battery Manufacturing





GenAI Supports the Factory of the Future

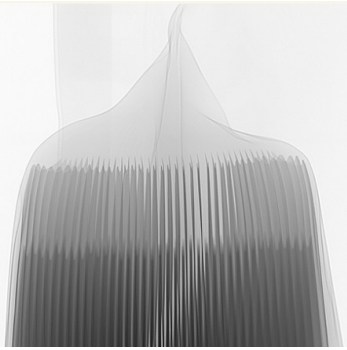
Generative AI's Role in the Factory of the Future | BCG

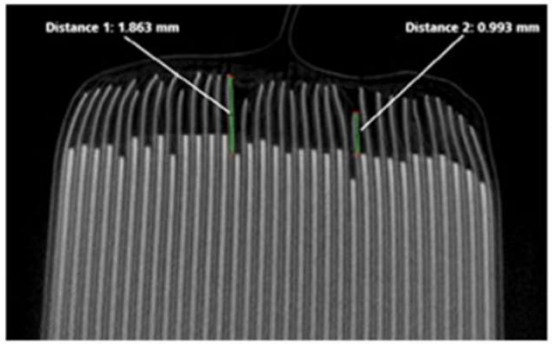
AI Transforms the 5M1E Framework into a Self-Optimizing System

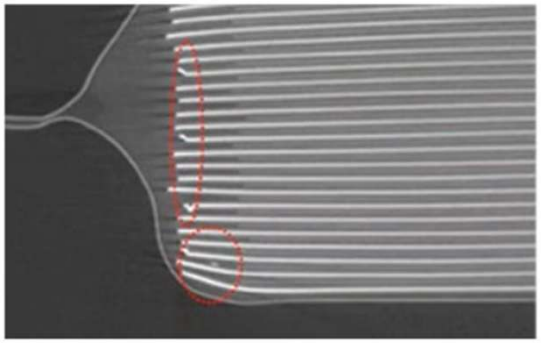
AI connects the 5M1E from data to decision

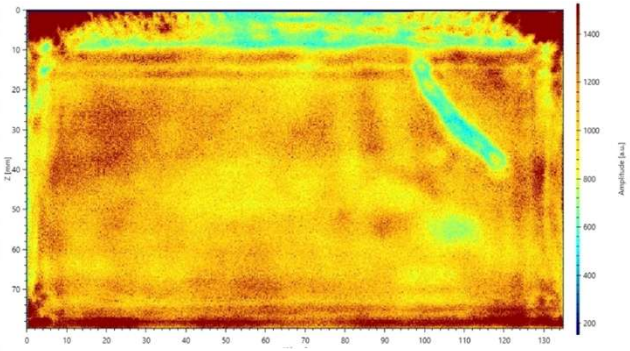
GenAI translates intelligence into human-centered action

5M1E	AI Transformation	Result
Man	GenAI copilots, digital SOPs, human–machine collaboration	Empowered, data-driven workforce
Machine	Predictive maintenance, adaptive control	Higher uptime, optimized yield
Material	Traceability via AI vision, predictive supply chain	Lower waste, stable flow
Method	Digital twins, AI optimization, GenAI process design	Continuous improvement, agility
Measurement	Intelligent vision, data fusion, AI root cause analysis	Faster feedback, zero defect quality
Environment	Energy optimization, emission tracking, AI simulation	Sustainable, controlled operations









X-Ray

Detection of A/C overhang

Image credit: Comet X-ray

Industrial CT

High-resolution image for clear detection of overhang
Detection of bent electrodes and a metal particle

Image credit: ZEISS

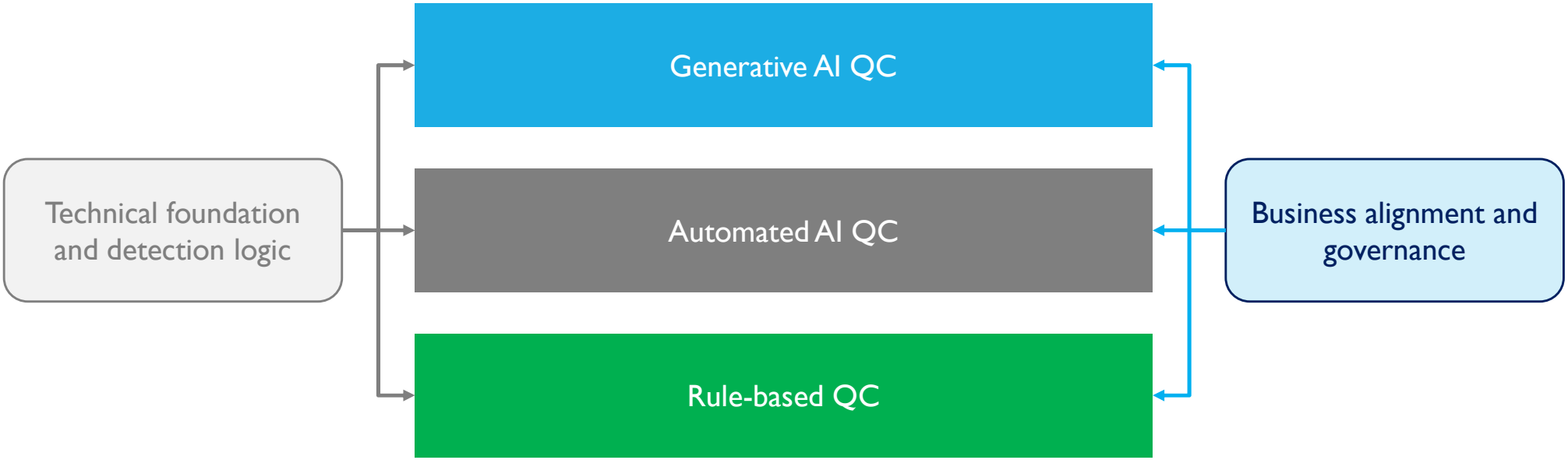
Ultrasound

Inspection of Wetted, Formed, and Aged Cells
Cell's electrolyte distribution & gas presence

Image credit: XARION

Case Highlights: AI EoL Detection

Predictive defect detection model, trained on full set of cell data to catch flawed cells.
Explainable, root cause analysis. Touch-free, no additional capex for QC devices



Capabilities progress bottom-up: from rules → automation → generative intelligence

Quality Control + AI

QC Approaches in Battery Manufacturing

Daniel Li | ManufAgent Solutions | EV & BATTERY FACTORIES FORUM 2026

Evolution

AI-Powered Agile Lean Manufacturing

Yesterday:

Efficiency and stability

Market trend:

Customized

High Mix Low Volume

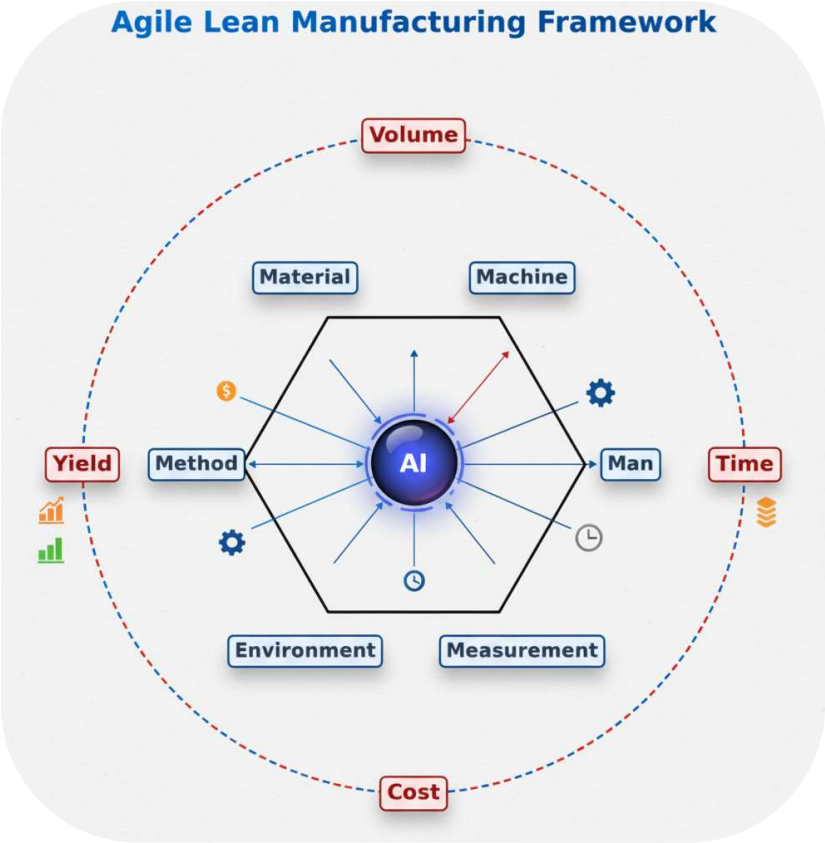
Today:

Agility and intelligence

AI powers real-time adaptation

4 Measurable KPIs: TVYC

Time, Volume, Yield, Cost



5M1E:
Man
Machine
Material
Method
Measurement
Environment

ManufAgent
integrates
Machine, Method,
Measurement, and
Environment

5M1E → MMM:
Man
Material
ManufAgent

AI Transforms 5M1E → MMM

Human + AI Collaboration = Agile Lean Manufacturing

Humans innovate/govern,

AI orchestrates/executes

Together they adapt, learn, and evolve

ManufAgent amplifies human capability

🏭 **ManufAgent**

=

🚗 **Autonomous Car**

ManufAgent *is to manufacturing what the autonomous car is to mobility*

Gigafactories that can learn to excel

Thank You

Daniel Li
ManufAgent Solutions